



$$\frac{dN_A}{dt} = -\frac{N_A}{\tau} \rightarrow N_A(t) = N_{A0} e^{-t/\tau}$$

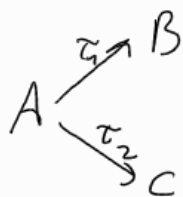
Razpadni zakon

Če $\tau \rightarrow t_{1/2}$ (razpadni čas):

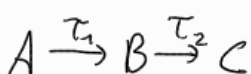
$$N_A(t) = N_{A0} 2^{-t/t_{1/2}} \Rightarrow t_{1/2} = \tau \ln 2$$

A... Aktivnost $[Bq = 1 \frac{\text{razpad}}{s}]$

$$A = \left| \frac{dN}{dt} \right|$$

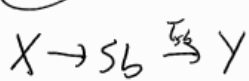


$$\frac{dN_A}{dt} = -\frac{N_A}{\tau_1} - \frac{N_A}{\tau_2}$$



$$\frac{dN_B}{dt} = \underbrace{\frac{N_A}{\tau_1}}_{?} - \frac{N_B}{\tau_2}$$

5.13 Zorko



je možno ko upr.
z razp. v X.

$$A_x = 3,5 \cdot 10^9 Bq = \text{konst.}$$

$$\tau_{sb} = 87 \text{ dni}$$

$$A_{sb}(0) = 1,2 \cdot 10^9 Bq$$

$$A_{sb}(t_0) = 2,5 \cdot 10^9 Bq$$

$$t_0 = ?$$

$$A_{sb \text{ max}} = ?$$

$$A_{sb}(t) = \frac{N_{sb}(t)}{\tau_{sb}}$$

$$\frac{dN_{sb}}{dt} = \underbrace{\frac{N_x}{\tau_x}}_{A_x} - \frac{N_{sb}}{\tau_{sb}}$$

$$\frac{dN_{sb}}{dt} + \frac{N_{sb}}{\tau_{sb}} = A_x$$

hom. del:

$$\frac{dN_{sb}}{dt} + \frac{N_{sb}}{\tau_{sb}} = 0 \Rightarrow N_{sb}^h(t) = K_h e^{-t/\tau_{sb}}$$

part. del:

$$\text{nast. } N_{sb}^p = K_p \rightarrow 0 + \frac{K_p}{\tau_{sb}} = A_x \rightarrow K_p = A_x \tau_{sb}$$

$$N_{sb}(t) = N_{sb}^h + N_{sb}^p = K_h e^{-t/\tau_{sb}} + A_x \tau_{sb}$$

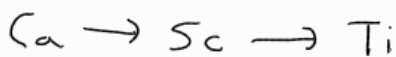
$$A_{sb}(t=0) = \frac{K_h + A_x \tau_{sb}}{\tau_{sb}} \rightarrow K_h = A_{sb}(0) \tau_{sb} - A_x \tau_{sb}$$

$$A_{sb}(t=0) = (A_{sb}(0) - A_x) e^{-t/\tau_{sb}} + A_x$$

$$\rightarrow t_0 = -\tau_{sb} \ln \frac{A_{sb}(t_0) - A_x}{A_{sb}(0) - A_x} = \underline{72 \text{ dni}}$$

$$A_{sb \max} = \lim_{t \rightarrow \infty} A_{sb}(t) = A_x$$

5.14 Zorko



$$t_{1/2}(Ca) = 4,8 \text{ dni} \rightarrow \tau_{Ca} = 6,92 \text{ dni}$$

$$t_{1/2}(Sc) = 3,43 \text{ dni} \rightarrow \tau_{Sc} = 4,95 \text{ dni}$$

$$m_{Ca} = 5 \text{ mg (ob } t=0 \text{ samo Ca)}$$

$$M(Ca) = 47 \text{ g/mol}$$

$$A(t_0) = \max.$$

$$A(t=0) = ?$$

$$t_0 = ?$$

$$A(t_0) = ?$$

$$A(t=0) = \frac{N_{Ca}(0)}{\tau_{Ca}} \frac{m_{Ca}}{M_{Ca}} N_A = \underline{1,07 \cdot 10^{14} \text{ Bq}}$$

$$A(t) = \frac{N_{Ca}}{\tau_{Ca}} + \frac{N_{Sc}}{\tau_{Sc}}$$

$$A_{Ca} = \frac{N_{Ca}(0)}{\tau_{Ca}} e^{-t/\tau_{Ca}}$$

$$\frac{dN_{Sc}}{dt} = \frac{N_{Ca}}{\tau_{Ca}} - \frac{N_{Sc}}{\tau_{Sc}}$$

$$K_p = A_0 \left(\frac{1}{\tau_{Sc}} - \frac{1}{\tau_{Ca}} \right)^{-1}$$

$$\rightarrow N_{Sc}(t) = K_h e^{-t/\tau_{Sc}} + \frac{A_0}{\frac{1}{\tau_{Sc}} - \frac{1}{\tau_{Ca}}} e^{-t/\tau_{Ca}}; N_{Sc}(0) = 0$$

$$K_h + \frac{A_0}{\frac{1}{\tau_{Sc}} - \frac{1}{\tau_{Ca}}} = 0$$

$$\text{hom: } N_{Sc}^h = K_h e^{-t/\tau_{Sc}}$$

$$\text{part.1 } N_{Sc}^p = K_p e^{-t/\tau_{Ca}}$$

$$= \frac{K_p}{\tau_{Ca}} e^{-t/\tau_{Ca}} + \frac{K_p}{\tau_{Sc}} e^{-t/\tau_{Ca}} = \frac{N_{Ca}(0)}{\tau_{Ca}} e^{-t/\tau_{Ca}}$$

$$K_p \left(\frac{1}{\tau_{Sc}} - \frac{1}{\tau_{Ca}} \right) = \frac{N_{Ca}(0)}{\tau_{Ca}}$$

$$\rightarrow N_{sc}(t) = \frac{A_0}{\frac{1}{\tau_{sc}} - \frac{1}{\tau_{cn}}} (e^{-t/\tau_{cn}} - e^{-t/\tau_{sc}})$$

$$A(t) = A_0 e^{-t/\tau_{cn}} + \frac{A_0}{1 - \frac{\tau_{sc}}{\tau_{cn}}} (e^{-t/\tau_{cn}} - e^{-t/\tau_{sc}}) = N_{cn}(0) \left[\underbrace{\left(\frac{1}{\tau_{cn}} + \frac{1}{\tau_{cn} - \tau_{sc}} \right)}_{C_1} e^{-t/\tau_{cn}} - \underbrace{\frac{1}{\tau_{cn} - \tau_{sc}}}_{C_2} e^{-t/\tau_{sc}} \right]$$

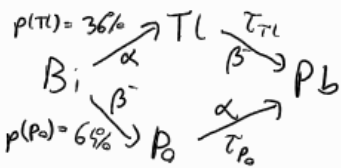
$$\left. \frac{dA}{dt} \right|_{t=t_0} = 0 = N_{cn}(0) \left[-\frac{C_1}{\tau_{cn}} e^{-t_0/\tau_{cn}} + \frac{C_2}{\tau_{sc}} e^{-t_0/\tau_{sc}} \right]$$

$$\frac{C_1}{C_2} \frac{\tau_{sc}}{\tau_{cn}} = \frac{e^{-t_0/\tau_{sc}}}{e^{-t_0/\tau_{cn}}} = e^{-t_0/\tau_{sc} + t_0/\tau_{cn}} / \ln$$

$$\frac{t_0}{\tau_{cn}} - \frac{t_0}{\tau_{sc}} = \ln \frac{C_1}{C_2} \frac{\tau_{sc}}{\tau_{cn}}$$

$$t_0 = \frac{1}{\frac{1}{\tau_{cn}} - \frac{1}{\tau_{sc}}} \ln \frac{C_1}{C_2} \frac{\tau_{sc}}{\tau_{cn}} = \underline{1,47 \text{ dni}} \rightarrow \underline{A(t_0) = 1,11 \cdot 10^{14} \text{ Bq}}$$

5.16



$$\tau_{Bi} = 87,3 \text{ min}$$

$$\tau_{Po} = 5,0 \cdot 10^{-9} \text{ min}$$

$$\tau_{Tl} = 4,5 \text{ min}$$

$$m_{Bi}(0) = 1 \mu\text{g}$$

$$M_{Bi} = 208,9804 \text{ g/mol}$$

$$A(t=1h) = ?$$

$$\frac{A_{\alpha}(t_0)}{A(t_0)} = ? \text{ (kak prispevanje } \alpha \text{ razpadi)}$$

$$A(t_0)$$

$$A(t) = A_{Bi}(t) + A_{Tl}(t) + A_{Po}(t)$$

$$A_{Bi}(t) = \frac{N_{Bi}(0)}{\tau_{Bi}} e^{-t/\tau_{Bi}}$$

$$\left. \frac{dN_{Tl}}{dt} = p(Tl) \frac{N_{Bi}}{\tau_{Bi}} - \frac{N_{Tl}}{\tau_{Tl}} \right| \left. \frac{dN_{Po}}{dt} = p(Po) \frac{N_{Bi}}{\tau_{Bi}} - \frac{N_{Po}}{\tau_{Po}} \right|$$